SMOLVLA / LIBERO | ADAPTIVE REPLANNING | 48 MATCHED EPISODES

Can the model decide when to replan?

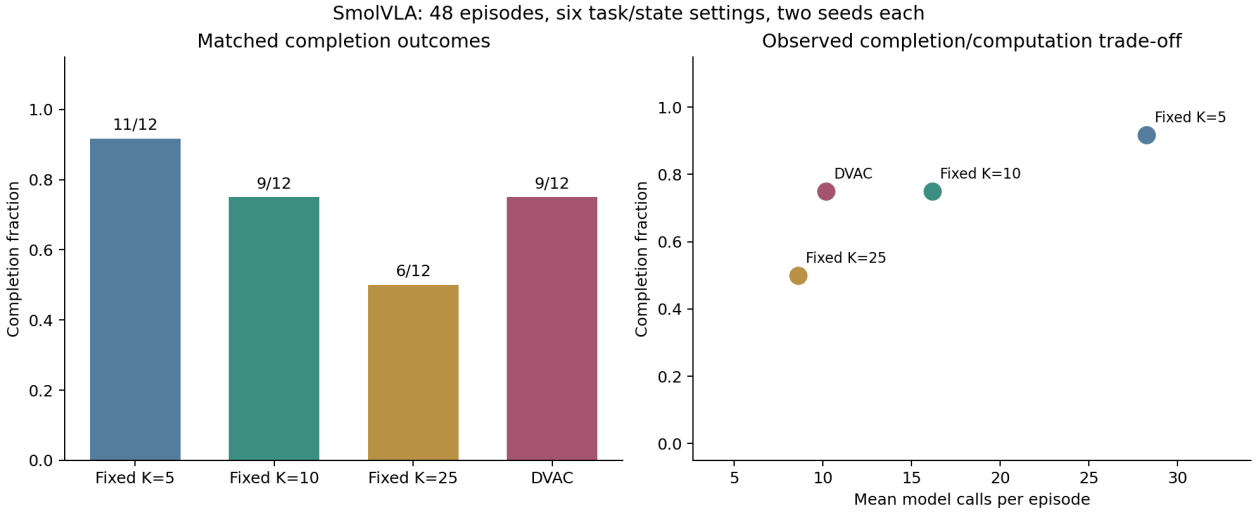
A denoising-variance decision rule compared with fixed execution intervals

A robot predicts a block of future actions, executes a prefix, observes the scene again and replans. This experiment tests whether the model's internal flow-matching trajectory can choose that prefix without additional model evaluations. The target is task completion with fewer policy calls.

DVAC completed 9/12 episodes. The fixed controls completed between 6 and 11/12. DVAC used 122 policy calls in total, averaging 10.2 calls per episode. All outcomes are retained.

Method	Completed	Calls total	Calls / action	Compute s / episode	Wall s / episode
Fixed K=5	11/12	339	0.203	26.19	30.68
Fixed K=10	9/12	194	0.102	17.29	21.74
Fixed K=25	6/12	103	0.042	12.73	17.64
DVAC	9/12	122	0.062	12.70	16.91

Calls/action is pooled over executed actions. Successful episodes may stop earlier, so call totals and wall time also depend on episode duration. Compute time includes instrumented inference, DVAC selection and simulator stepping.



Each point uses 12 matched task/state/seed episodes. There are six physical task/state settings, each repeated with two sampling seeds. These observations do not establish a general benchmark ranking.

From flow matching to an execution horizon

H=50 future actions are generated in M=10 internal Euler updates. The robot executes K actions before receiving its next observation. Internal iteration i and future action index j are different axes.

1. Generate the complete action block

The input c contains both camera images, robot state and the instruction. Starting from Gaussian noise X(0), the model updates the entire action block. The input c stays fixed; the simulator is paused.

$$t_i = 1 - i/M,\ v_i = v_\theta(X^{(i)}, t_i, c)$$
$$X^{(i+1)} = X^{(i)} - v_i/M,\ i = 0,\ldots,9$$

2. Measure consistency inside that one generation

Capture a clean-action estimate before each Euler update. Keep the final L=5 estimates with iteration set I={5,6,7,8,9} (flow times approximately 0.5, 0.4, 0.3, 0.2, 0.1). For each future action j, compute population variance across estimates and sum over seven valid action coordinates:

$$Z_i = X^{(i)} - t_i v_i$$
$$\mu_{j,d} = (1/L)\ \Sigma_{i\in I}\ Z_{i,j,d}$$
$$V_j = \Sigma_{d=0..6}\ (1/L)\ \Sigma_{i\in I}\ (Z_{i,j,d} - \mu_{j,d})^2$$

The estimates concern the same future action inside one call. Normalized action channels are used and padding is excluded. Float32 estimates are reduced in float64.

3. Choose a prefix using the preceding calls

W contains all 50 variance values from each of up to five preceding calls, including repeated values. Add the current call only after selecting K. Use population standard deviation.

$$\tau = \text{mean}(W) + 2\ \text{std}_{pop}(W)$$
$$j^* = \text{first } j \text{ with } V_j > \tau,\ j = 0,\ldots,49$$
$$K = 50 \text{ if no crossing exists; otherwise } K = \max(1, j^*)$$

Execute the first K actions from the final X(10), with the original postprocessing. For example, j*=4 executes indices 0-3. The minimum K=1 can force execution even when index 0 crosses. The mean estimate is never used as a control command.

Paper defaults used: L=5, history=5, alpha=2, minimum=1, maximum=H. Explicit adaptation choices: reset history each episode; use K=1 and no threshold for its first call; analyze seven valid SmolVLA channels. The paper does not prescribe the empty-history rule. [1]

Matched design and complete outcomes

Three tasks x two initial states (15, 16) x two policy seeds (11, 22) x four methods = 48 episodes. The six physical settings produce 12 task/state/seed blocks. Every block contains all four methods; block order and method order are randomized with seed 20261024.

All methods use the same checkpoint, 50-action horizon, ten solver iterations, FP32 precision, 20 Hz control, ten settling steps and two warmup calls. Soup and bowl episodes allow 280 actions; drawer episodes allow 300. The RTX 2060 runs methods sequentially. Parameters were fixed before collection.

Task	State	Seed	Fixed 5	Fixed 10	Fixed 25	DVAC
Soup into basket	15	11	S / 28	S / 14	S / 5	S / 11
Soup into basket	15	22	S / 31	S / 16	S / 6	S / 7
Soup into basket	16	11	S / 30	S / 13	S / 5	S / 5
Soup into basket	16	22	S / 29	S / 15	S / 7	F / 23
Bowl onto plate	15	11	S / 15	S / 8	F / 12	S / 8
Bowl onto plate	15	22	S / 15	S / 8	S / 4	S / 3
Bowl onto plate	16	11	S / 17	F / 28	F / 12	S / 8
Bowl onto plate	16	22	S / 16	S / 8	S / 4	S / 4
Open middle drawer	15	11	S / 25	S / 12	F / 12	S / 7
Open middle drawer	15	22	F / 60	F / 30	F / 12	F / 16
Open middle drawer	16	11	S / 28	S / 12	F / 12	S / 7
Open middle drawer	16	22	S / 45	F / 30	F / 12	F / 23

Every cell is outcome / model calls; S = success, F = failure. These are all 12 paired blocks, not selected examples. Initial simulator-state and observation hashes match across all methods and both seeds within each physical setting.

Instrumentation preserves the sampled actions

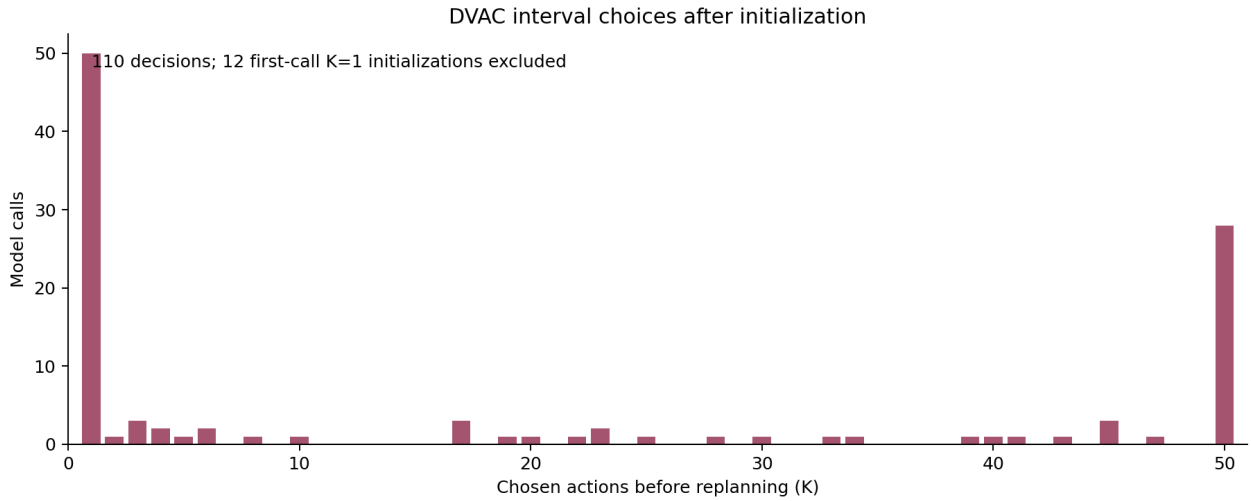
Before collection, plain and instrumented inference were run on excluded initial state 21, with sampling seed 101, for all three tasks. Every final 50 x 7 action value matched exactly. The wrapper observes existing velocity evaluations and returns each original tensor unchanged.

Held-out task	Final action arrays	Maximum absolute error
Soup into basket	Exactly equal	0
Bowl onto plate	Exactly equal	0
Open middle drawer	Exactly equal	0

The independent verifier replayed 758 generated chunks and 7995 executed actions: variance, thresholds, chosen horizons, clipped prefixes, seeds and raw-file hashes. First complete predictions also match across all methods within each state/seed block. Exact equality on these inputs is an integrity check, not a proof of universal numerical equivalence.

What execution intervals did DVAC choose?

Across all calls, mean K was 17.78, median 1, and range 1-50. The actual executed prefix averaged 16.17 actions because success or termination may interrupt a selected prefix.



The histogram excludes the 12 required first-call K=1 choices. Among 110 later decisions, 50 chose K=1 and 28 chose K=50. There were 28 calls without any strict crossing.

Matched outcome differences

Reference	DVAC only succeeds	Fixed only succeeds	Both succeed	Both fail	Mean call difference
Fixed K=5	0	2	9	1	-18.1
Fixed K=10	1	1	8	2	-6.0
Fixed K=25	4	1	5	2	+1.6

Call differences are DVAC minus the reference on the same 12 blocks. Seeds share physical initial situations; no p-value or independence-based significance claim is made.

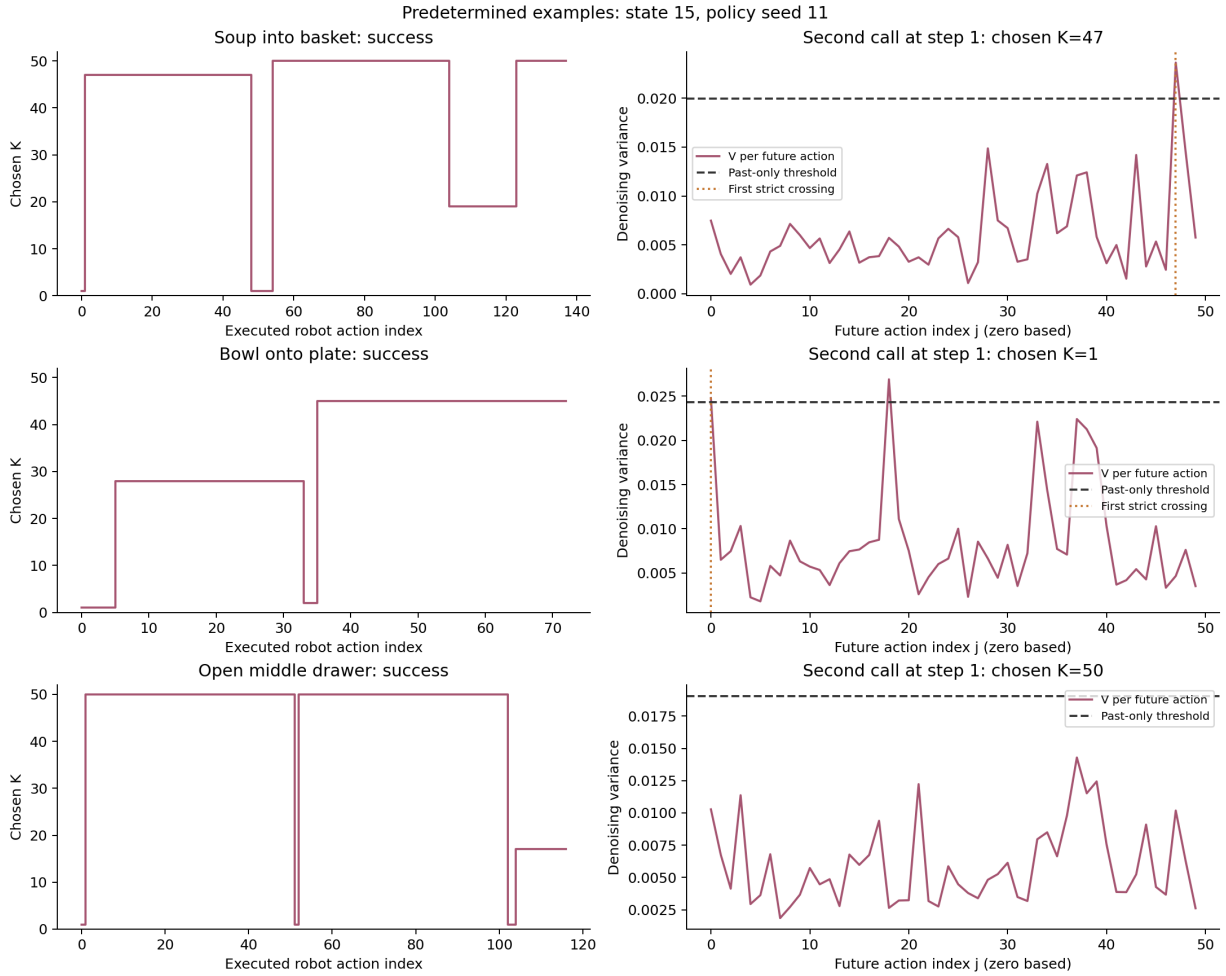
Completion by task

Task	Fixed 5	Fixed 10	Fixed 25	DVAC
Soup into basket	4/4	4/4	4/4	3/4
Bowl onto plate	4/4	3/4	2/4	4/4
Open middle drawer	3/4	2/4	0/4	2/4

Relative to fixed K=10, DVAC's completion difference was +0.0 percentage points and its mean call difference was -6.0. The measured selector cost was 0.049 ms per DVAC call; instrumented inference averaged 0.803 s per call. These are observed sample means.

Three examples selected before the outcomes

Each task is shown at initial state 15 and policy seed 11. The left panels trace selected execution horizons against robot action index. The right panels always inspect the second policy call, immediately after the first-call initialization.



A horizontal threshold is computed from earlier calls. A vertical marker identifies the first strict crossing, when one exists. The final action trajectory is generated before this prefix decision. The plots reveal the decision rule; they do not identify the physical cause of a failure.

Task	Outcome	Second-call K	Threshold	Maximum V
Soup into basket	Success	47	0.0199	0.0236
Bowl onto plate	Success	1	0.0243	0.0269
Open middle drawer	Success	50	0.0191	0.0143

The companion recording, outputs/dvac-report/demo.mp4, shows these same three task/state/seed pairs: fixed K=10 on the left and DVAC on the right, aligned by simulation time. It includes instructions, robot motion, replanning markers, selected K and final outcomes. All three preselected pairs are included regardless of success.

Interpretation, limits and reproducibility

In this sample, DVAC completed 9/12 episodes, compared with 9/12 for fixed K=10, and used fewer policy calls. Its internal signal therefore needs to be assessed jointly with the observed outcomes and computational cost. A stable denoising trajectory may still produce an incorrect action; large variance is not a calibrated probability of failure.

An outcome-selected failure example

In the outcome-selected soup-can example (state 16, seed 22), sampled frames show the can remaining on the table while the DVAC arm moves toward the basket. At steps 3 and 53, DVAC selected K=50: maximum V was 0.0301 and 0.0140, below thresholds 0.0435 and 0.0388. Later K=1 calls near the basket did not recover completion. The matched fixed-10 run visibly carries the can to the basket and succeeds in 143 actions; DVAC times out at 280. This is consistent with a missed initial grasp, but the sampled views do not identify the contact-mechanical cause or prove that another interval would have corrected that same trajectory.

Assistant inspection of sampled frames; no independent human review. Source frames and uncertain observations are retained in outputs/dvac-report/review/.

Scope of the evidence

Only two initial states per task and two sampling seeds per state were tested. Six physical situations are too few for broad generalization. The Wilson intervals in the analysis files are descriptive and do not adjust for seed dependence. This study transfers the DVAC rule to SmolVLA; it does not reproduce the paper's full benchmark or establish an official LIBERO score.

The comparison combines fresh observations and stochastic regeneration. It does not isolate which mechanism caused a success difference. The variance scale depends on the checkpoint, normalization, solver grid and estimate window. Gripper close-command counts and path length are control diagnostics, not verified grasp-error labels. Neither the plots nor aggregate outcomes justify a causal claim about grasping or motion correction.

Timing definitions

All four methods collect identical diagnostics. Inference includes the policy prediction and variance reduction; selector time is separate. Compute is inference + selection + simulation. Episode wall time includes reset, rollout and video closure, excluding JSON writing and hashing. The simulator pauses during inference. This is not a real-time latency or asynchronous control study. The ordered equality checks are unsuitable for estimating instrumentation overhead.

Reproduce from the repository

```
bash scripts/collect_dvac.sh check configs/dvac.json \
 --output outputs/dvac-check/invariance.json
bash scripts/collect_dvac.sh --config configs/dvac.json --output outputs/dvac
python scripts/analyze_dvac.py outputs/dvac outputs/dvac-report --decode-videos
python scripts/build_dvac_demo.py
python scripts/build_dvac_report.py
```

Use the pinned Linux collection environment and the artifact environment for analysis/PDF tooling. Render the generated PDF and inspect every page before delivery. The manifest preserves checkpoint/backbone revisions, runtime, hardware and sampler hashes. The PDF's adjacent JSON binds its statistics, figures, source files and raw evidence to SHA-256 checksums.

```
Checkpoint: HuggingFaceVLA/smolvla_libero
Revision: 6721902bc4d61e50a3bfdb11dfb4cb626f05d102
Statistics SHA-256: 54318c302e340e726e696a18deef25fd0aab91496062584ec7ac777ace4a6dba
```

References

1. Denoising Tells When to Replan: Denoising-Variance Adaptive Chunking for Flow-Based Robot Policies. Section 3 and Algorithm 1.

2. SmolVLA: A Vision-Language-Action Model for Affordable and Efficient Robotics.

3. Frozen SmolVLA LIBERO checkpoint.

4. LeRobot 0.6.1 SmolVLA implementation.